
Piecewise Rules for Rates and Fees

Writing piecewise fee rules, reading a cost back to its input, finding a missing rate, and comparing two schedules.

Every price list below charges one rate up to a breakpoint and a different rate after it, so no single linear equation describes the whole cost. Work each one this way:

- Name the input variable and its unit (hours, pounds, kilowatt-hours, miles, gigabytes, pages) and name the output — always a total cost in dollars.
- Write the rule branch by branch, each branch with the interval of inputs it governs. Use the form $C(x) = \text{flat fee} + (\text{rate})(x - \text{breakpoint})$ for any branch that starts charging only past the breakpoint.
- Decide which branch the question lands in *before* you compute.

1. A city garage charges \$4.00 for a stay of up to 2 hours, and \$3.00 for each additional hour after the first 2, for stays up to 8 hours. Let t be the number of hours parked and $C(t)$ the charge in dollars. Write the piecewise rule for $C(t)$, then find the charge for a 5-hour stay.

Answer: _____

2. A shipper charges a flat \$6.00 for any package weighing up to 3 pounds, and \$6.00 plus \$1.25 for each pound over 3 pounds beyond that. Let w be the package weight in pounds and $C(w)$ the charge in dollars. Write the piecewise rule for $C(w)$, then find the charge for a 7-pound package.

Answer: _____

3. An electric utility bills the first 500 kilowatt-hours of a month at \$0.11 per kilowatt-hour and every kilowatt-hour above 500 at \$0.16 per kilowatt-hour. Let n be the kilowatt-hours used and $C(n)$ the bill in dollars. Priya used 800 kilowatt-hours and says her bill is \$128.00. Write the piecewise rule for $C(n)$, explain in one sentence what Priya did wrong, then compute the correct bill.

Answer: _____

4. A taxi charges \$3.50 for the first mile and \$2.25 for each additional mile. Let m be the trip length in miles and $C(m)$ the fare in dollars. Write the piecewise rule for $C(m)$, then find the trip length that produces a fare of exactly \$17.00.

Answer: _____ mi

5. A community pool charges \$8.00 for the first 3 hours of a visit and the same unknown amount k dollars for each additional hour. A 7-hour visit costs \$22.00. Write the piecewise rule for the cost of a visit of t hours using k , then find the value of k .

Answer: _____

6. A phone plan costs \$30.00 per month and includes 5 gigabytes of data. Each gigabyte above 5 costs \$12.00, but the monthly bill is never allowed to exceed \$90.00. Let g be the gigabytes used and $C(g)$ the bill in dollars. Write the three-branch piecewise rule for $C(g)$, then find the bill for a month in which 9 gigabytes were used.

Answer: _____

7. Using the same phone plan as problem 6 (\$30.00 base, 5 gigabytes included, \$12.00 per gigabyte above 5, bill capped at \$90.00), find the data usage at which the middle branch first reaches the \$90.00 cap — that is, the breakpoint where the flat capped branch takes over.

Answer: _____ GB

8. Look again at the phone-plan rule from problems 6 and 7. Explain why the middle branch and the capped branch must give the same cost at the point where they meet, and describe what a customer would see on the bill if the two branches disagreed there by a few dollars. Write in complete sentences.

9. Two garages sit on the same block. Garage A charges \$4.00 for a stay of up to 2 hours and \$3.00 for each additional hour. Garage B charges \$2.50 per hour from the moment you arrive, with no flat fee. For a stay longer than 2 hours, let t be the hours parked and y the cost in dollars. Write a two-equation system for the stay on which the two garages cost the same, then solve it for t and y .

Answer: _____

10. A print shop charges a \$25.00 setup fee plus \$0.40 per page for the first 200 pages of a job, and a reduced rate of r dollars per page for every page beyond 200. A 500-page job costs \$153.00. Write the piecewise rule for the cost of a p -page job using r , then find r .

Answer: _____